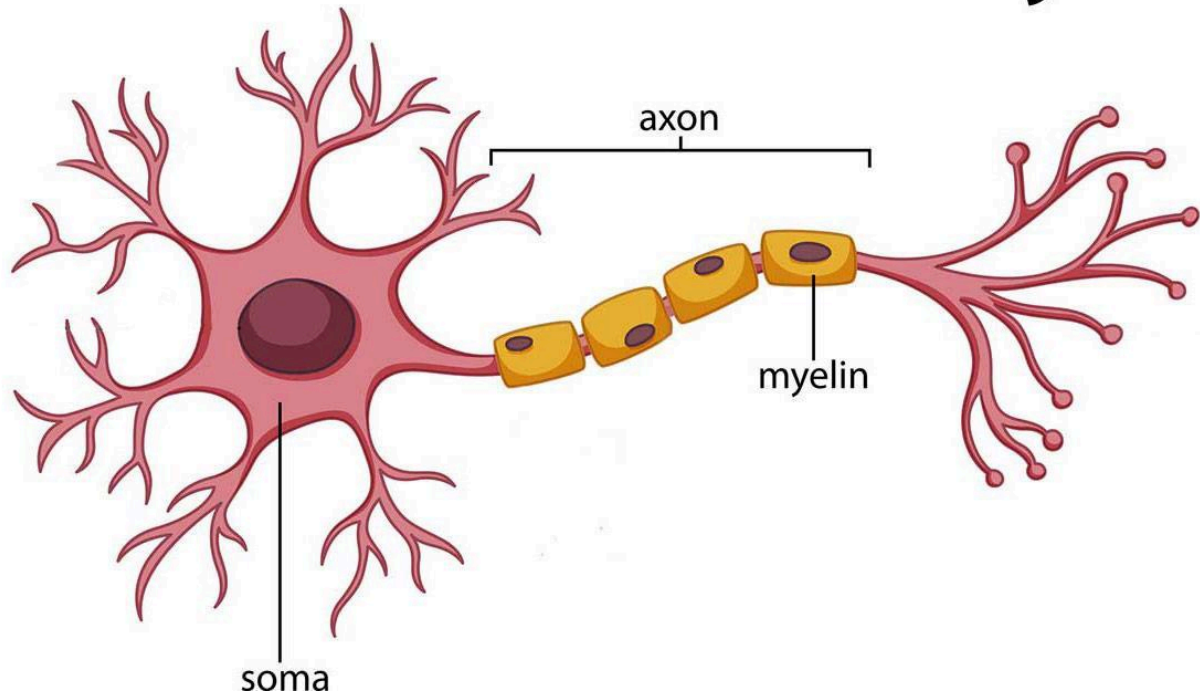
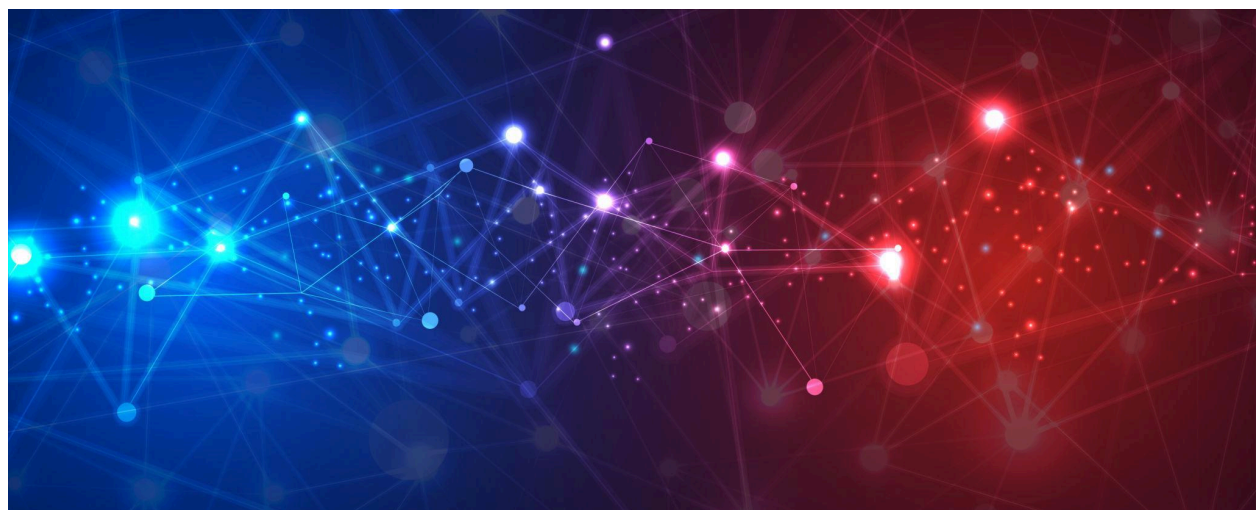
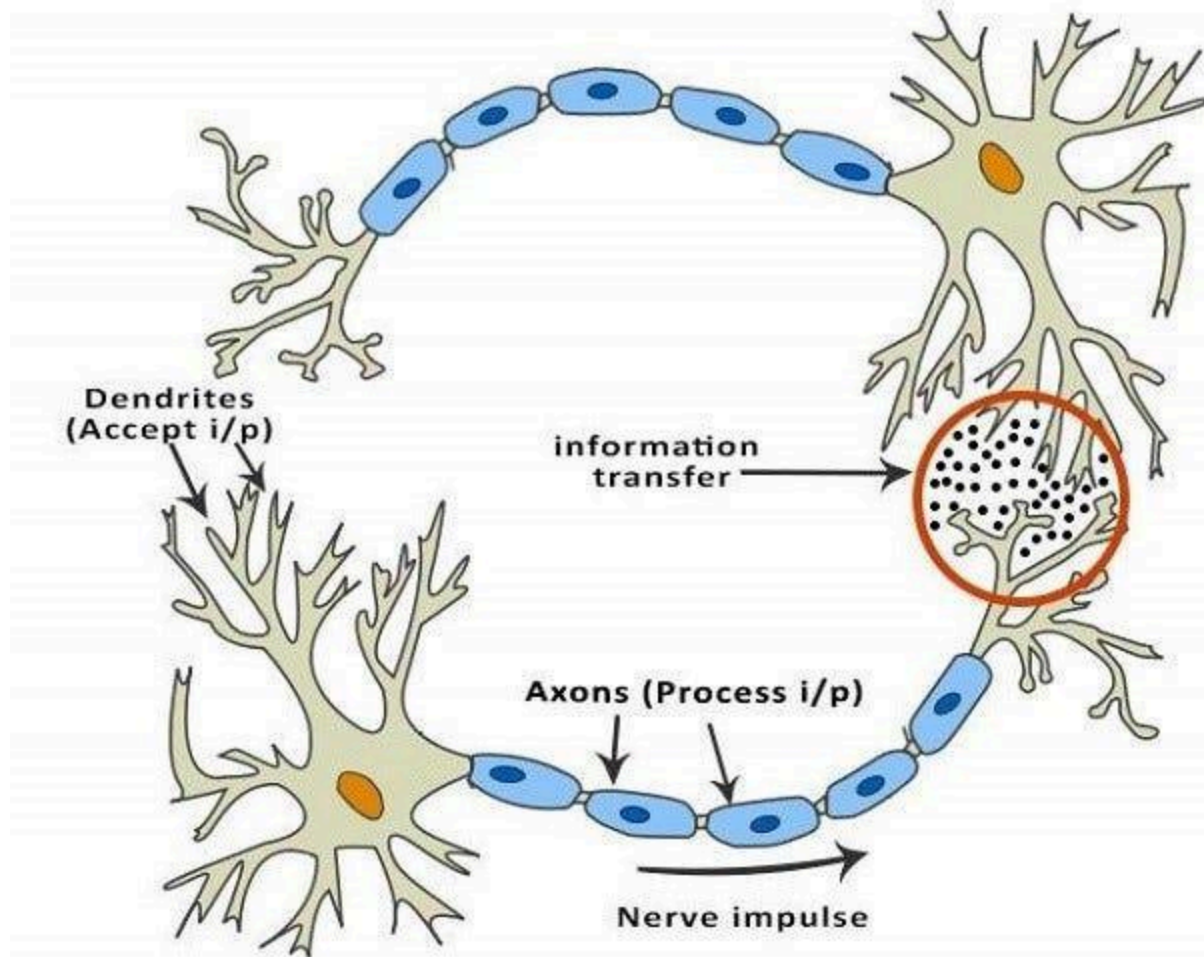


1. Human Brain Neural Network (First Image)

Neuron Anatomy

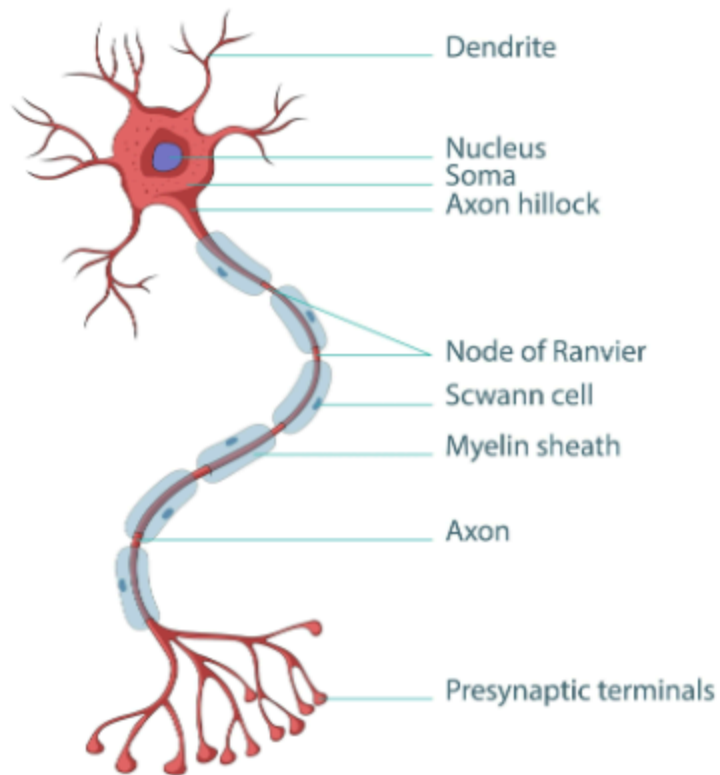




5

This image explains how a **human brain neuron** works.

NEURON



Parts of Human Neuron

Part	Simple Meaning	Work
Dendrites	Input receiver	Receives information
Cell Body	Processing center	Processes information
Axon	Output path	Sends information
Nerve Impulse	Signal/message	Transfers data to next neuron

=====

Simple Working

Step 1:

Dendrites receive information.

Example:

Touching hot object

Step 2:

Cell body processes the signal.

Brain understands:

"Hot object!"

Step 3:

Axon sends signal quickly.

Step 4:

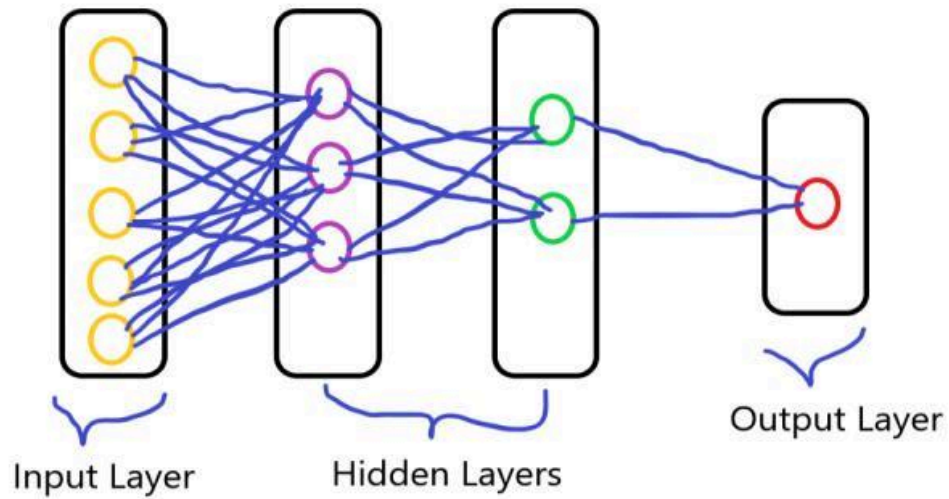
Body reacts.

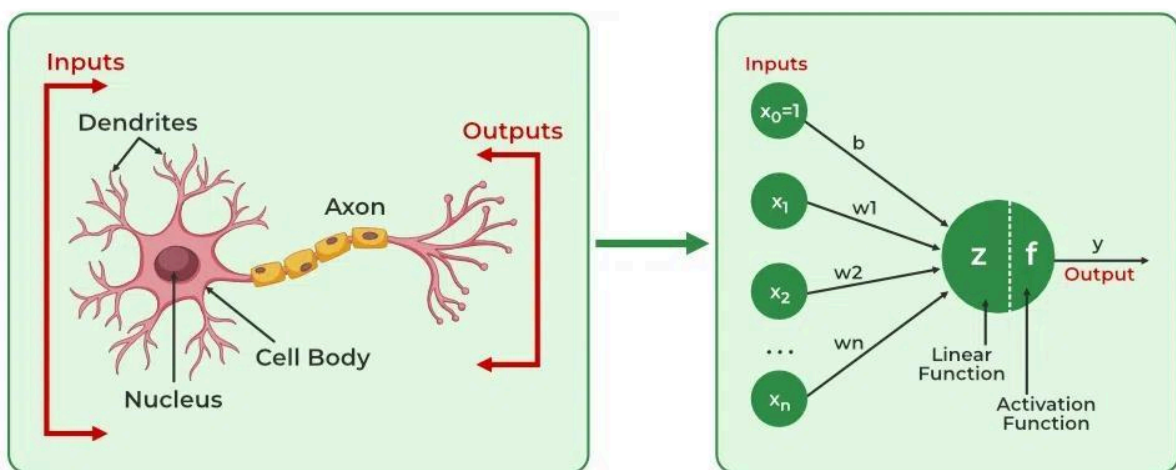
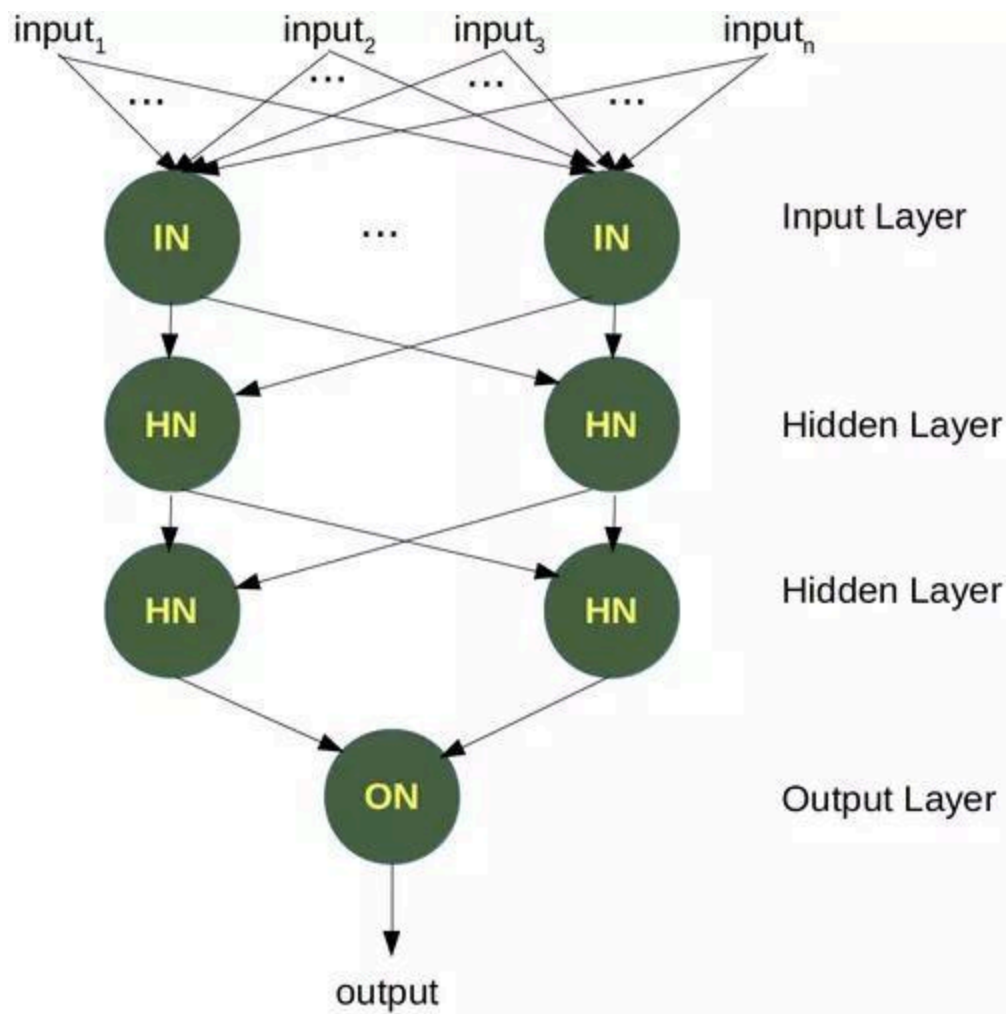
Example:

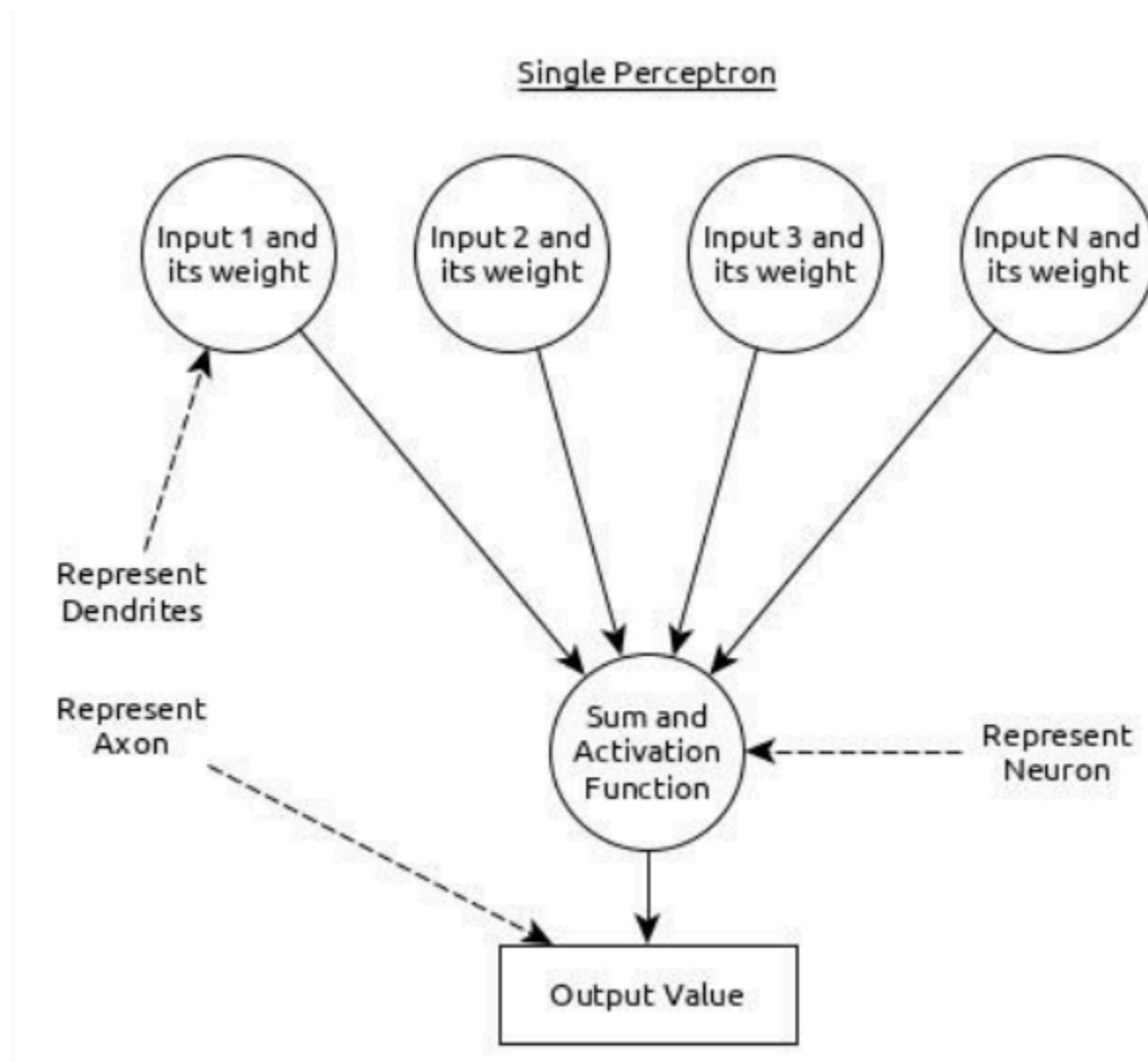
Hand moves away

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2. Artificial Neural Network (ANN) (Second Image)







This is a computer version inspired by the human brain.

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ANN Components Explained Simply

Component

Meaning

Work

Input Layer	Receives data	Takes input
Weight	Importance value	Decides importance
Hidden Layer	Processing layer	Learns patterns
Activation Function	Decision maker	Decides output
Loss Function	Error checker	Finds mistakes

=====

Simple Example

Problem:

Predict whether student passes or fails.

Input Layer Receives:

Study Hours = 8
Attendance = 90%
Marks = 75

Weight

Important values get more focus.

Example:

Marks may be more important than attendance

Hidden Layer

Learns patterns from previous student data.

Activation Function

Makes final decision.

Pass or Fail

Loss Function

Checks prediction error.

Example:

Predicted = Pass

Actual = Fail

Then system learns and improves.

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Human Brain vs Artificial Neural Network

Human Brain	Artificial Neural Network
Dendrites receive signals	Input layer receives data
Brain processes information	Hidden layer processes data
Axon sends output	Output layer gives result
Learns from experience	Learns from training data

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Easy One-Line Understanding

Human brain neurons inspired the creation of Artificial Neural Networks in Deep Learning.

